

PAYMENT DETAILS MANAGEMENT

1. Database Design:

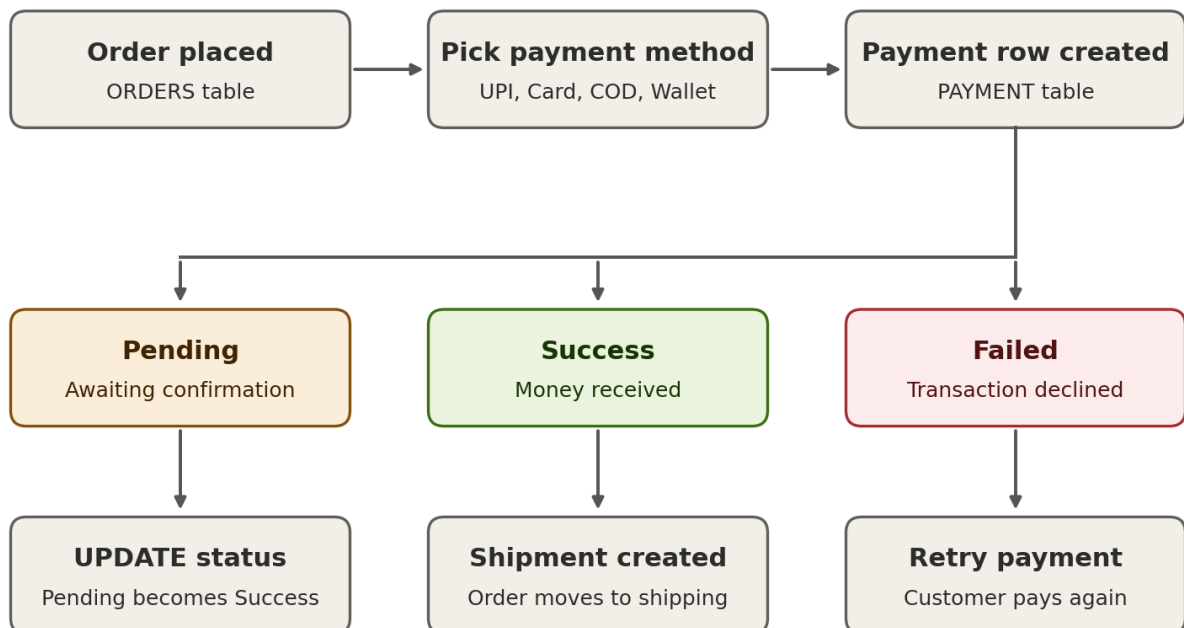
1.1 Payment Table

Attributes of Payment Table

Attribute	Description	Key / Constraint
PAYMENT_ID	Unique identification number of the payment	Primary Key
ORDER_ID	Identifies the order associated with the payment	Foreign Key, Not Null
PAYMENT_METHOD	Mode used for making the payment	Not Null
PAYMENT_STATUS	Status of the payment transaction	Not Null, Check
PAYMENT_DATE	Date on which the payment was made	Default SYSDATE
AMOUNT	Amount paid for the order	Not Null, Check (> 0)

2. Table Relationships

Each order has exactly one payment (One-to-One, total participation on both sides). The flow below shows how an order reaches the PAYMENT table and what happens for each payment status.



3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE PAYMENT (  
 2     PAYMENT_ID      NUMBER PRIMARY KEY,  
 3     ORDER_ID        NUMBER NOT NULL,  
 4     PAYMENT_METHOD  VARCHAR2(30) NOT NULL,  
 5     PAYMENT_STATUS  VARCHAR2(20) NOT NULL,  
 6     PAYMENT_DATE    DATE DEFAULT SYSDATE,  
 7     AMOUNT          NUMBER(10,2) NOT NULL,  
 8     CONSTRAINT fk_pay_order FOREIGN KEY (ORDER_ID)  
 9         REFERENCES ORDERS(ORDER_ID),  
10     CONSTRAINT chk_pay_status  
11         CHECK (PAYMENT_STATUS IN ('Success','Pending','Failed')),  
12     CONSTRAINT chk_pay_amount CHECK (AMOUNT > 0)  
13 );
```

Table created.

3.2 Describe Payment Table

```
SQL> DESC PAYMENT  
Name                               Null?    Type  
-----  
PAYMENT_ID                         NOT NULL NUMBER  
ORDER_ID                           NOT NULL NUMBER  
PAYMENT_METHOD                     NOT NULL VARCHAR2(30)  
PAYMENT_STATUS                     NOT NULL VARCHAR2(20)  
PAYMENT_DATE                       DATE  
AMOUNT                             NOT NULL NUMBER(10,2)
```

4. Payment Data Insertion

```
SQL> INSERT INTO PAYMENT VALUES  
 2  (401,201,'UPI','Success',TO_DATE('10-09-2026','DD-MM-YYYY'),1299);  
  
1 row created.  
  
SQL> INSERT INTO PAYMENT VALUES  
 2  (402,202,'Credit Card','Success',TO_DATE('11-09-2026','DD-MM-YYYY'),2499);  
  
1 row created.  
  
SQL> INSERT INTO PAYMENT VALUES  
 2  (403,203,'Debit Card','Success',TO_DATE('12-09-2026','DD-MM-YYYY'),899);  
  
1 row created.  
  
SQL> INSERT INTO PAYMENT VALUES  
 2  (404,204,'UPI','Pending',TO_DATE('13-09-2026','DD-MM-YYYY'),1799);  
  
1 row created.  
  
SQL> INSERT INTO PAYMENT VALUES  
 2  (405,205,'Cash on Delivery','Pending',TO_DATE('14-09-2026','DD-MM-YYYY'),999);  
  
1 row created.
```

```

SQL> INSERT INTO PAYMENT VALUES
  2  (406,206,'Net Banking','Success',TO_DATE('15-09-2026','DD-MM-YYYY'),3499);

1 row created.

SQL> INSERT INTO PAYMENT VALUES
  2  (407,207,'Credit Card','Failed',TO_DATE('16-09-2026','DD-MM-YYYY'),1599);

1 row created.

SQL> INSERT INTO PAYMENT VALUES
  2  (408,208,'Wallet','Success',TO_DATE('17-09-2026','DD-MM-YYYY'),699);

1 row created.

SQL> INSERT INTO PAYMENT VALUES
  2  (409,209,'UPI','Success',TO_DATE('18-09-2026','DD-MM-YYYY'),2199);

1 row created.

SQL> INSERT INTO PAYMENT VALUES
  2  (410,210,'Cash on Delivery','Success',TO_DATE('19-09-2026','DD-MM-YYYY'),1199);

1 row created.

SQL> COMMIT;

Commit complete.

```

5. Manage Payment Transactions

Display settings

```

SQL> SET LINESIZE 150
SQL> COLUMN PAYMENT_METHOD FORMAT A20
SQL> COLUMN PAYMENT_STATUS FORMAT A15

```

5.1 Display Successful Payments

```

SQL> SELECT PAYMENT_ID, ORDER_ID, PAYMENT_METHOD, PAYMENT_DATE, AMOUNT
  2  FROM PAYMENT
  3  WHERE PAYMENT_STATUS = 'Success';

```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_D	AMOUNT
401	201	UPI	10-SEP-26	1299
402	202	Credit Card	11-SEP-26	2499
403	203	Debit Card	12-SEP-26	899
406	206	Net Banking	15-SEP-26	3499
408	208	Wallet	17-SEP-26	699
409	209	UPI	18-SEP-26	2199
410	210	Cash on Delivery	19-SEP-26	1199

7 rows selected.

5.2 Display Failed Payments

```
SQL> SELECT PAYMENT_ID, ORDER_ID, PAYMENT_METHOD, PAYMENT_DATE, AMOUNT, PAYMENT_STATUS
2 FROM PAYMENT
3 WHERE PAYMENT_STATUS = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_D	AMOUNT	PAYMENT_STATUS
407	207	Credit Card	16-SEP-26	1599	Failed

1 row selected.

6. Modify Payment Status

```
SQL> UPDATE PAYMENT
2 SET PAYMENT_STATUS = 'Success'
3 WHERE PAYMENT_ID = 404;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT * FROM PAYMENT WHERE PAYMENT_ID = 404;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
404	204	UPI	Success	13-SEP-26	1799

1 row selected.

7. Analyze Payment Methods

```
SQL> SELECT PAYMENT_METHOD,
2 COUNT(*) AS TOTAL_TRANSACTIONS,
3 SUM(AMOUNT) AS TOTAL_AMOUNT
4 FROM PAYMENT
5 GROUP BY PAYMENT_METHOD
6 ORDER BY TOTAL_TRANSACTIONS DESC, TOTAL_AMOUNT DESC;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS	TOTAL_AMOUNT
UPI	3	5297
Credit Card	2	4098
Cash on Delivery	2	2198
Net Banking	1	3499
Debit Card	1	899
Wallet	1	699

6 rows selected.

8. Payment Transaction Report

```
SQL> SELECT PAYMENT_ID, ORDER_ID, PAYMENT_METHOD, PAYMENT_STATUS, PAYMENT_DATE, AMOUNT
2  FROM PAYMENT
3  ORDER BY PAYMENT_DATE;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
401	201	UPI	Success	10-SEP-26	1299
402	202	Credit Card	Success	11-SEP-26	2499
403	203	Debit Card	Success	12-SEP-26	899
404	204	UPI	Success	13-SEP-26	1799
405	205	Cash on Delivery	Pending	14-SEP-26	999
406	206	Net Banking	Success	15-SEP-26	3499
407	207	Credit Card	Failed	16-SEP-26	1599
408	208	Wallet	Success	17-SEP-26	699
409	209	UPI	Success	18-SEP-26	2199
410	210	Cash on Delivery	Success	19-SEP-26	1199

10 rows selected.